

A Configurable Surface-Electrode Ion Trap Design for Quantum Information Processing *

LIU Wei(刘威)^{1,2**}, CHEN Shu-Ming(陈书明)^{1,2**}, CHEN Ping-Xing(陈平形)³, WU Wei(吴伟)³

¹College of Computer, National University of Defense Technology, Changsha 410073

²Science and Technology on PDL, National University of Defense Technology, Changsha 410073

³College of Science, National University of Defense Technology, Changsha 410073

(Received 20 July 2013)

We propose a configurable surface-electrode ion trap design to alleviate the poor reusability of the existing traps. It can architecturally and electrically support 5 mainstream modes by design reuse, thus enhancing the trap reusability and reducing the experiment setup overhead. We also develop a corresponding simulation suite which can optimize trap geometries and calculate trap parameters to control the trapped ion's classic motion. According to our analytical and simulated results, the configurable design can serve as a unified platform for basic research of large-scale quantum information processing.

PACS: 37.10.Ty, 03.67.Lx, 41.20.-q

DOI: 10.1088/0256-307X/30/12/123702

According to DiVincenzo's criterion,^[1] ion trap is a leading candidate for quantum information processing systems.^[2] However, a remaining challenge is the scaling of trapped ions to tremendous qubits which are necessary for solving nontrivial problems.^[3] Surface-electrode (SE) ion trap is an effective scale approach due to its compatibility with the existing semiconductor process, thus a multi-trapping zone can be easily fabricated and connected.^[4,5] The past decade has seen many such chips classified into two types, symmetrical traps^[6-8] with electrodes symmetrical about the trap's axial direction and asymmetrical traps^[9,10] with unequal width of rf electrodes^[9] or unequal static dc electrodes.^[10]

Although making the rf electrodes unequal is a method^[9] to rotate the trap axis which is necessary for the Doppler cooling,^[11] this type of trap has two disadvantages. Firstly, the scalability is poor as the junctions seem too difficult to construct.^[12] Secondly, the axial controlling dc voltages is difficult to be calculated. Breaking the rf ground electrode (Fig. 1) into two halves is another method,^[6] while it makes the ion close to the insulator layer. Nowadays, the mainstream traps always adopt the same rf electrodes and a whole rf ground.^[3] They rotate the trap axes by unequal dc electrodes or unequal voltage amplitude.

The scalable capacity of mainstream SE ion traps is attractive,^[2] however the experiment is still subject to limited scale since there is no well accepted answer about which trap layout is optimal for scalability. Specifically, the existing traps^[6-10] can only support simplex operating mode for scalability, i.e. their reusability is poor. Moreover, such simplex increases the setup overhead of each experiment as the vacuum requirement. In order to alleviate these problems, it is urgent to design a unified and configurable platform, based on which one can evaluate and compare many

modes quickly, then choose the optimum mode.

In this Letter, we propose a configurable SE ion trap design (Fig. 1) and an associated software suite for optimizing and calculating the trap parameters. As shown in Fig. 1, the proposed design can architecturally (unequal dimension of the dc electrodes) and electrically (unequal amplitude of applied control voltages) support symmetric and asymmetric ion trap operation by design reuse, thus significantly shorten the time of experiment setup. The software suite is developed based on firstly the quantitative analysis to determine the length and the width of dc electrodes, then the integration of COMSOL (a finite element method based software) and MATLAB to iteratively simulate and optimize the trap depth, calculate the ion axial transport controlling voltages and so on.

As shown in Fig. 1, there are two rf electrodes applied voltage $V_{\text{rf}} \cos(\Omega t)$. When other electrodes are grounded, they can create a pseudopotential field with a local minimum above the x - z plane. Such minimum is the trap center where the trapped ions oscillate. The dimension parameters a and b determine the position of the trap center and the trap depth.^[13-16] Specifically, the trap center locates at $x_0 = a/2$, $y_0 = \sqrt{2ab + a^2}/2$, and the trap depth is the pseudopotential at the saddle point:

$$\psi_E = \frac{eV_{\text{rf}}^2}{\pi^2 m \Omega^2} \left[\frac{b}{(a+b)^2 + (a+b)\sqrt{2ab + a^2}} \right]^2,$$

where e and m are the charge and mass of the trapped ion. In this study, we just restate these extraordinary conclusions to determine the dimension of the rf electrodes.

About the dc electrode design, existing studies adopt the hypothesis that the dc electrodes are semi-infinite in the x direction,^[13-16] while all traps are fabricated with finite dimension. Moreover, there are no

*Supported by the National Natural Science Foundation of China under Grant No 61205108, and the High Performance Computing (HPC) Foundation of National University of Defense Technology.

**Correspondence authors. Email: wliu@nudt.edu.cn; smchen@nudt.edu.cn

© 2013 Chinese Physical Society and IOP Publishing Ltd

studies before to determine the length of the dc electrodes. Next, we quantitatively analyze such paradox, then give rules for practical dc electrode fabrication. The potential of a rectangular electrode is expressed by

$$\phi(x, y, z) = \frac{V}{2\pi} \left\{ \arctan[(x_2 - x)(z_2 - z)/y\sqrt{p_2}] \right. \\ \left. - \arctan[(x_1 - x)(z_2 - z)/y\sqrt{p_1}] \right. \\ \left. - \arctan[(x_2 - x)(z_1 - z)/y\sqrt{q_2}] \right. \\ \left. + \arctan[(x_1 - x)(z_1 - z)/y\sqrt{q_1}] \right\}; \quad (1)$$

$$\phi(x, 0, z) = \begin{cases} V, & x_1 < x < x_2 \cap z_1 < z < z_2, \\ 0, & \text{otherwise;} \end{cases} \quad (2)$$

$$p_n = y^2 + (x_n - x)^2 + (z_2 - z)^2, \\ q_n = y^2 + (x_n - x)^2 + (z_1 - z)^2, \quad (3)$$

where $n \in \{1, 2\}$. We emphasize on the ion's classical motion along the z direction, i.e. x and y in Eq. (1) are constant. Then $(x_1 - x)$ and $(x_2 - x)$ are the x coordinates of the left and right edges of the analyzed dc electrode; x_1 is always constant since it is adjacent to the rf electrode which has been fixed by a and b .

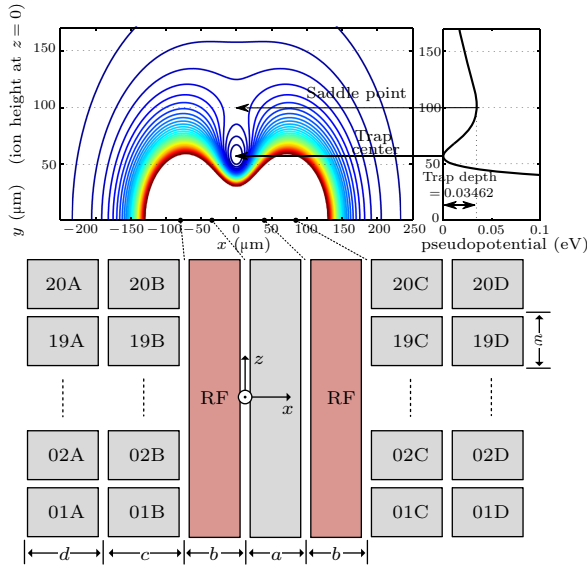


Fig. 1. The layout of the proposed configurable surface-electrode ion trap in the x - z plane. It consists of seven vertical electrode rails: four dc electrode rails named as A, B, C and D, two rf electrodes marked by rf and one rf ground in the middle. Each dc electrode rail is horizontally segmented into 20 sects numbered from 01 to 20. Particularly, 01 AB denotes a composite electrode and 01A, 01B denote two separate ones. The original point is illustrated above. The parameters a , b , c , d , and w define the dimensions of the electrodes. For the ion $^{40}\text{Ca}^+$, the top shows the pseudopotential in the x - y plane at $z = 0$ with $a = 80 \mu\text{m}$, $b = 40 \mu\text{m}$, $V_{\text{rf}} = 91 \text{ V}$, $\Omega/2\pi = 58.55 \text{ MHz}$.^[12] The trap depth is 0.03462 eV which is obtained by COMSOL. According to the equation of ψ_E , the analytical trap depth is 0.0303 eV .

In order to study the variety of potential ϕ influenced by the finite dc electrode length, an intuitive

method is to analyze the first derivative of ϕ with respect to x_2 . According to Eq. (1), we have

$$\frac{\partial \phi}{\partial x_2} = \frac{V}{2\pi} \left\{ \frac{y^3(z_2 - z) + y(z_2 - z)^3}{\sqrt{p_2}[y^2 p_2 + (x_2 - x)^2(z_2 - z)^2]} \right. \\ \left. - \frac{y^3(z_1 - z) + y(z_1 - z)^3}{\sqrt{q_2}[y^2 q_2 + (x_2 - x)^2(z_1 - z)^2]} \right\}. \quad (4)$$

Since $z_2 = w/2 = -z_1 > 0$, it can be easily proved that $\frac{\partial^2 \phi}{\partial x_2^2} < 0$. Thus Eq. (4) is a monotonically decreasing function. In addition, with x_2 increasing, the limitation of $\frac{\partial \phi}{\partial x_2}$ is zero according to Eq. (4). These two conclusions are consistent with Fig. 2. We can predict that, taking the electrode 02CD as an example, there should be a threshold length T_l for electrode 02CD, if its length exceeds T_l , then the extra electrode proportion will have a weak influence on the spatial potential.

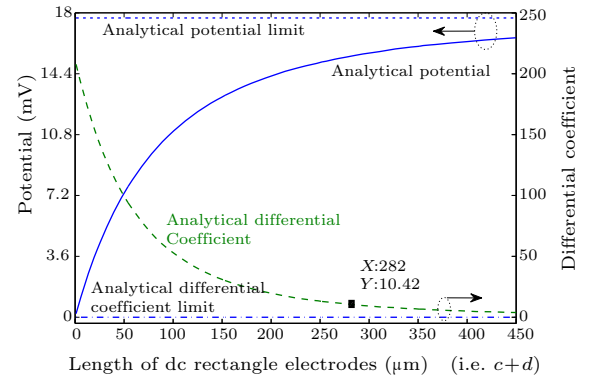


Fig. 2. Quantitative analysis of the potential ϕ located at $(50 \mu\text{m}, 81 \mu\text{m}, 0)$ with respect to the length of the rectangle electrode with diagonal coordinates $(180 \mu\text{m}, 0, -30 \mu\text{m})$ and $(x_2, 0, 30 \mu\text{m})$. The electrode is set to 1 V . The length of the analyzed rectangle electrode is $(x_2 - 180) \mu\text{m}$.

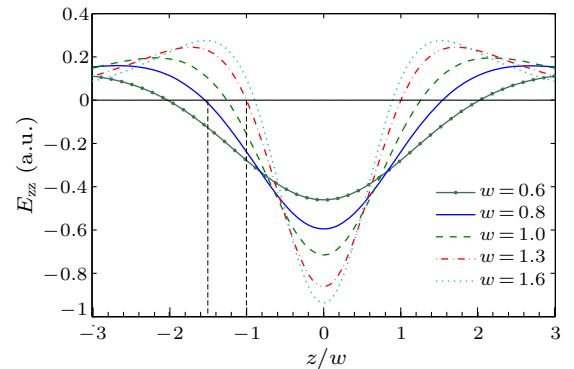


Fig. 3. The electric field curvature $E_{zz} = \partial^2 \phi / \partial z^2$ in z direction at the trap center with respect to z/w for one electrode applied to 1 V . Here $z = 0$ is aligned with the center of the electrode. The other dimension parameters are $a = 1$, $b = 1.2$, $(c + d) = 2$, thus the trap center is at $x_0 = 0.5$, $y_0 = 0.92$. As w increases, the field curvature significantly changes near the center of the electrode and the zero points of the five potential curvatures move toward to the original point.

As shown in Fig. 2, the limits of ϕ and $\frac{\partial \phi}{\partial x_2}$ are

17.7 mV and 0, respectively. The first-steep-then-smooth variation is consistent with the prediction that there should be a threshold length. We define T_l as the length when the maximum differential coefficient decreases to its δ . For example, $T_l = 282 \mu\text{m}$ when $\delta = 5\%$, as shown in Fig. 2. The parameter δ is determined by the variation of the potential and the practical area of the substrate jointly. Thus, the parameter T_l can be a helpful criteria to determine the actual length of the dc electrode in ion trap fabrication.

In the same way, the variety of ϕ influenced by a rectangle electrode's width can also be quantitatively analyzed through the function $\frac{\partial\phi}{\partial z_2}$. The whole variation trend is the same as that in Fig. 2, while there are some differences. Only when z_2 is very large, ϕ and $\frac{\partial\phi}{\partial z_2}$ reach their limits, respectively. For example, applying 1 V to a electrode diagonally located at $(180 \mu\text{m}, 0, -z_2)$ and $(380 \mu\text{m}, 0, z_2)$, and analyzing the potential at $(50 \mu\text{m}, 81 \mu\text{m}, 0)$. When $\frac{\partial\phi}{\partial z_2}$ decreases to its 5%, z_2 needs to be $500 \mu\text{m}$, i.e. the width threshold T_w is $1000 \mu\text{m}$ which is much larger than T_l in Fig. 2.

The parameter T_w is too coarse to guide the fabrication. We propose two more elaborate design criteria to determine the parameter w . The primary motivation of segmenting dc electrodes is to trap ions in the z direction and to move the trapped ion. For criteria 1, in the interest to axially transport an ion under the heating constraint, we have to maintain the curvature of the potential constant in the z direction.^[13] For criteria 2, in another interest to achieve high ion density, we set a single potential well which occupies the width of three consecutive dc electrodes. This is because $3w$ is the minimum width of a single trapping well.

Table 1. The characteristics of the configurable modes supported by the proposed layout. Modes 1–4 configure the length in the x direction to architecturally (Arch.) and electrically (Elec.) support symmetrical (sym) and asymmetrical (asym) experiments.

	seg-dc rail	Arch.	Elec.	θ
Mode 1	AB & CD	sym	sym & asym	$-\pi/2 - \pi/2$
Mode 2	B & C	sym	sym & asym	$-\pi/2 - \pi/2$
Mode 3	AB & C	asym	sym	fixed
Mode 4	B & CD	asym	sym	fixed
Mode 5	configure the width of a single trapping well in z direction			

Figure 3 shows E_{zz} for five dc electrode widths. According to the criteria 1, the constant potential curvature means that the corresponding electric field should have significant curvature everywhere within the influence scope of a electrode. The influence scope is defined by the distance of the two zero points in each field curvature. In Fig. 3, criteria 1 is satisfied for all the five w . According to the criteria 2, the effective influence scope of a electrode must extend from $2w$ to $3w$. In other words, only the field curvature with the zero points ranking from ± 1.5 to ± 1 is eligible. The width selected here is smaller than the recommended in Ref. [13] because it neglected in our proposed criteria 2. In brief, only the electrodes with

width $0.8 < w \leq 1.3$ satisfy the two criteria.

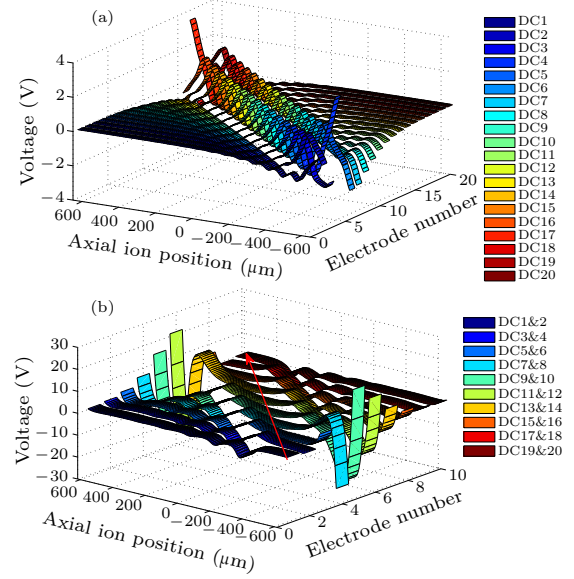


Fig. 4. Voltage trajectories for ion transport plotted as ribbons against the axial trapping position. (a) The width of the 20 dc electrodes is $80 \mu\text{m}$ corresponding to the $w = 0.8$ in Fig. 3. Other dimensions are set the same as the way in Fig. 3. A clear periodicity in axial ion position and the trap center can be clearly observed. Ion is transported from $-650 \mu\text{m}$ to $650 \mu\text{m}$. (b) The 10 dc electrodes are made up with the immediate neighbor electrodes, i.e. its width is $160 \mu\text{m}$. The red arrow shows the trap center and the ion is transported from $-610 \mu\text{m}$ to $610 \mu\text{m}$. Since the composite electrodes are wider, the voltage valley is wider and the voltage amplitude is larger. Such width is undesirable.

The proposed configurable design is based on the above analytical conclusions. Such design can support 5 modes which are depicted in Table 1. These modes give emphasis to the configuration of the dc electrodes' length and width. Modes 1–4 configure the length in the x direction to architecturally (Arch.) and electrically (Elec.) support symmetrical (sym) and asymmetrical (asym) experiments. Mode 5 configures the width in the z direction to change a single well's dimension.

Mode 1 resembles a symmetric “five-wire” trap. According to Eq. (4), since the dc rails, A and D, are playing a role in this mode, the length of electrode rail B must be shorter than the threshold T_l . We set the length of the rail A to be equal to T_l , then we can obtain $T < c + d < 2T$. Thus, electrode AB can maximize the influence of the spatial potential with a minimum voltage. In this mode, the trap axes are rotated by applying appropriate asymmetrical static voltages to rail AB and CD. The rotation angle depends on the applied voltages, and is typically in the range of $-\pi/2 \leq \theta \leq \pi/2$.

Mode 2 is a seven-wire trap. The middle five electrode rails act as mode 1. Ion transporting is achieved by applying slowly varying voltages to electrodes in rails B and C. Then the remained rails A and D can be used for rotation of axes and micromotion com-

pensation solely and synchronously. This is another reason why we set the length of A and D equal to T_l . The rails A and D reside far from the rf electrodes, so they have weaker influence on the rf field. We can apply larger opposite voltages to rails A and D, thus a variable rotation angle can be achieved easier.

Modes 3 and 4 are asymmetrical. The remained electrode rail A or D needs to be grounded. The trap axes is rotated by applying the same voltage to electrode rails of unequal length, like AB and C or B and CD. As the ration of the two unequal length is fixed, the rotation angle is also fixed.

Mode 5 configures the width of a single well through combining two adjacent dc electrodes in the z direction, for example, 01A and 02A, or using more than 3 pairs of dc electrodes to construct a single well. It has two motivations: firstly, analyzing the crosstalk effect^[16] between two neighboring wells. Secondly, analyzing the ion axially transport influenced by the parameter w . These effects must be studied in detail for a scalable SE ion trap containing many ions in many wells. In the x direction, mode 5 supports any one of the modes 1–4.

Until now, we have introduced the proposed hardware in detail. However, for a complete configurable chip design, an associated software suite is necessary. In the following, taking the ion transport in mode 5 as an example, we depict how the developed software is used to calculate the different configurable voltages.

Transporting the ion is a “backward” problem in which the potential of all the trap centers are prior determined, then the voltages for dc electrodes are needed to be simulated and calculated iteratively. To obtain a general solution to Laplace’s equation $\Delta\phi_i=0$, we expand the potential in terms of spherical harmonics coefficients up to order 2. Then we obtain the equation^[17]

$$\begin{aligned} \frac{\phi_i(x, y, z)}{V} &= \frac{1}{V} \left[M_{i,1} + M_{i,2} \left(\frac{y}{r_0} \right) + M_{i,3} \left(\frac{z}{r_0} \right) \right. \\ &\quad + M_{i,4} \left(\frac{x}{r_0} \right) + M_{i,5} \left(\frac{xy}{2r_0^2} \right) + M_{i,6} \left(\frac{zy}{2r_0^2} \right) \\ &\quad + M_{i,7} \left(\frac{2z^2 - x^2 - y^2}{2r_0^2} \right) + M_{i,8} \left(\frac{zx}{2r_0^2} \right) \\ &\quad \left. + M_{i,9} \left(\frac{x^2 - y^2}{2r_0^2} \right) + O(r^3) \right] \\ &= \sum_{j=1}^9 M_{ji} Y_j + O(r^3), \end{aligned} \quad (5)$$

where i represents the i th one of the N dc electrodes. M_{ji} are unitless spherical harmonics coefficients. The potential $\phi_i(x, y, z)$ is simulated by applying 1 V to the i th dc electrode. According to the superposition principle, we can obtain $\phi = \sum_{i=1}^N \sum_{j=1}^9 M_{ji} V_i Y_j$. The coefficients can be written in a matrix equation

$\widetilde{M}V=M$, where $\widetilde{M}=\{M_{ji}\}$, $V=\{V_i\}$, $M=\{M_j\}$ and $j \in \{1, 2, \dots, 9\}$. Setting a specific M_j to 1 V and the others are 0 V, then the dc voltages needed to form a single trap center can be calculated through a matrix inversion operation.^[18]

Figure 4 shows the results simulated by the proposed software suitable for $M_7 = 1$ V, taking mode 1 as an example. Symmetrically, Fig. 4 only shows the voltages applied to the right side electrodes in Fig. 1. The voltage “valleys” located at the trap centers, as shown clearly in Fig. 4(a). In Fig. 4(b), it is marked by an arrow. We define L as the length between the two pinnacles of the valley. The L in Fig. 4(a) is 242 μm , and in Fig. 4(b) is 480 μm , which are corresponding to the configuration of $w = 0.8$ and $w = 1.6$ in Fig. 3. This is because our single well’s width is $3w$. On the other hand, the change amplitude of voltages outside the valley in Fig. 4(a) is flatter than that in Fig. 4(b). This is because the influence scope of $w = 1.6$ is smaller than that of $w = 0.8$, thus larger voltages are needed to provide stronger influence for distant positions. These conclusions are consistent well with that of Fig. 3 and our proposed criteria 2.

In conclusion, we have proposed a configurable ion trap design with detailed theory analysis. For the electrode dimension, we have proposed some design criteria to achieve design reuse. The trap platform can support 5 operation modes. These modes improve the reusability of the existing ion traps and reduce the experiment setup overhead. Additionally, we develop a corresponding software suitable to validate the design criteria and simulate the voltages for controlling the classic motion of a single ion. Thus our configurable design can serve as the basis platform for a large-scale ion trap in quantum information processing.

References

- [1] DiVincenzo D P 2000 *Fortschr. Phys.* **48** 771
- [2] Ladd T D et al 2010 *Nature* **464** 45
- [3] Monroe C and Kim J 2013 *Science* **339** 1164
- [4] Chiaverini J et al 2005 *Quantum Inf. Comput.* **5** 419
- [5] Amini J M, Britton J, Leibfried D and Wineland D J 2011 *Atom Chips* (Germany: Wiley-VCH) p 395
- [6] Leibbrandt D R et al 2007 *Phys. Rev. A* **76** 055403
- [7] Allcock D T C et al 2010 *New J. Phys.* **12** 053026
- [8] Bowler R et al 2012 *Phys. Rev. Lett.* **109** 080502
- [9] Seidelin S et al 2006 *Phys. Rev. Lett.* **96** 253003
- [10] Doret S C et al 2012 *New J. Phys.* **14** 073012
- [11] Eschner J et al 2003 *J. Opt. Soc. Am. B* **20** 1003
- [12] Wright K et al 2013 *New J. Phys.* **15** 033004
- [13] House M G 2008 *Phys. Rev. A* **78** 033402
- [14] Nizamani A H and Hensinger W K 2012 *Appl. Phys. B* **106** 327
- [15] Ji W B et al 2011 *Chin. Phys. Lett.* **28** 073701
- [16] Ji W B et al 2012 *Chin. Phys. B* **21** 063701
- [17] Berkeland D J et al 1998 *J. Appl. Phys.* **83** 5025
- [18] Littich G 2011 *Master Thesis* (University of California)
- [19] Ben-Israel A and Greville Thomas N E 2010 *Generalized Inverses* (Berlin: Springer) p 93